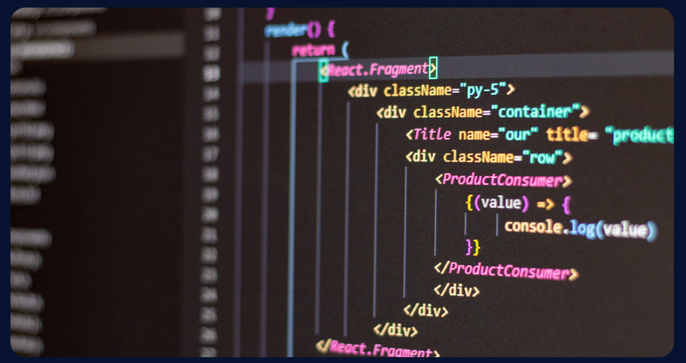

應用python於海洋環境資料分析

Presented By

謝宜臻(Yi-Chen,Hsieh)

前言INTRODUCTION

近幾年來全球氣候變遷變得更加嚴重，珊瑚白化問題也更加複雜，也因為研究氣候變遷是當要之務，再加上對珊瑚的影響有強大的興趣，想瞭解這幾年來的變化和緣由，所以未來的論文應該會往這方面繼續鑽研下去，想藉由這次的報告開始

In recent years, global climate change has become more severe, making the issue of coral bleaching increasingly complex.

Given the urgency of studying climate change and a strong interest in understanding its impact on corals, there is a desire to explore the changes and causes observed in recent years. Therefore, future research papers will likely delve deeper into this area. This report serves as a starting point for such investigations.

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"Happy Holidays! BCO-DMO will be on break from 23 December to 2 January 2025. Submissions and questions will still be accepted, however our responses may be delayed during this time."

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DATABASE

Programs	53
Projects	1,683
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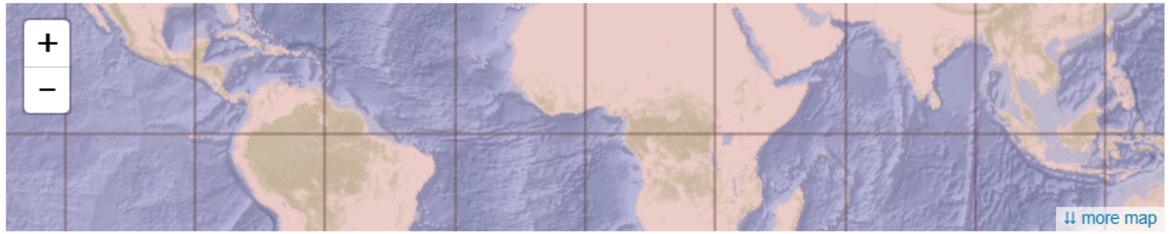
Platforms	673
Datasets	6,941
Publications	5,408
Instruments	687
Parameters	1,599
People	4,021
Affiliations	748
Funding	115

Dataset: Global Bleaching and Environmental Data

Cite This Dataset

DOI:10.26008/1912/bco-dmo.773466.2



Spatial Extent: ***N****:36.75* ***E****:179.964472* ***S****:-30.2625* ***W****:-179.974333*

Temporal Extent: 1980-06-15 - 2020-08-15

Project: Identifying coral reef 'bright spots' from the global 2015-2017 thermal-stress event (Coral Reef Brightspots)

Principal Investigator: Robert van Woesik (Florida Institute of Technology, FIT)

Co-Principal Investigator: Deron Burkepile (University of California-Santa Barbara, UCSB)

- coral reef bleaching dataset website(1980~2020)

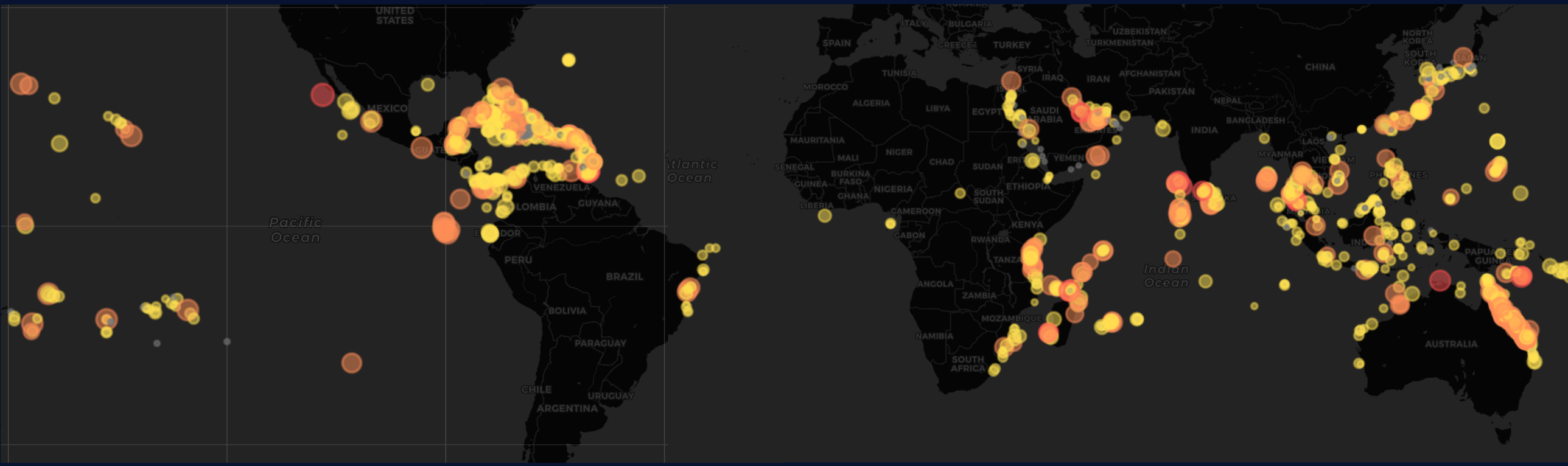
all dataset

```
  df=pd.read_csv('global_bleaching_environmental (1).csv', encoding='latin1')
       Site_ID  Sample_ID  ...                                      Sample_Comments  Bleaching_Comments
0         2501   10324336  ...                                                   nd                  nd
1         3467   10324754  ...  The bleaching does not  appear to have gained ...                  nd
2         1794   10323866  ...                                                   nd                  nd
3         8647   10328028  ...                                                   nd                  nd
4         8648   10328029  ...                                                   nd                  nd
...        ...        ...  ...                                                  ...                 ...
41356    15446   10310562  ...                                                   nd                  nd
41357    15456   10310527  ...                                                   nd                  nd
41358    15456   10310527  ...                                                   nd                  nd
41359    15457   10310536  ...                                                   nd                  nd
41360    15457   10310536  ...                                                   nd                  nd

[41361 rows x 62 columns]
```

There are 41360 data

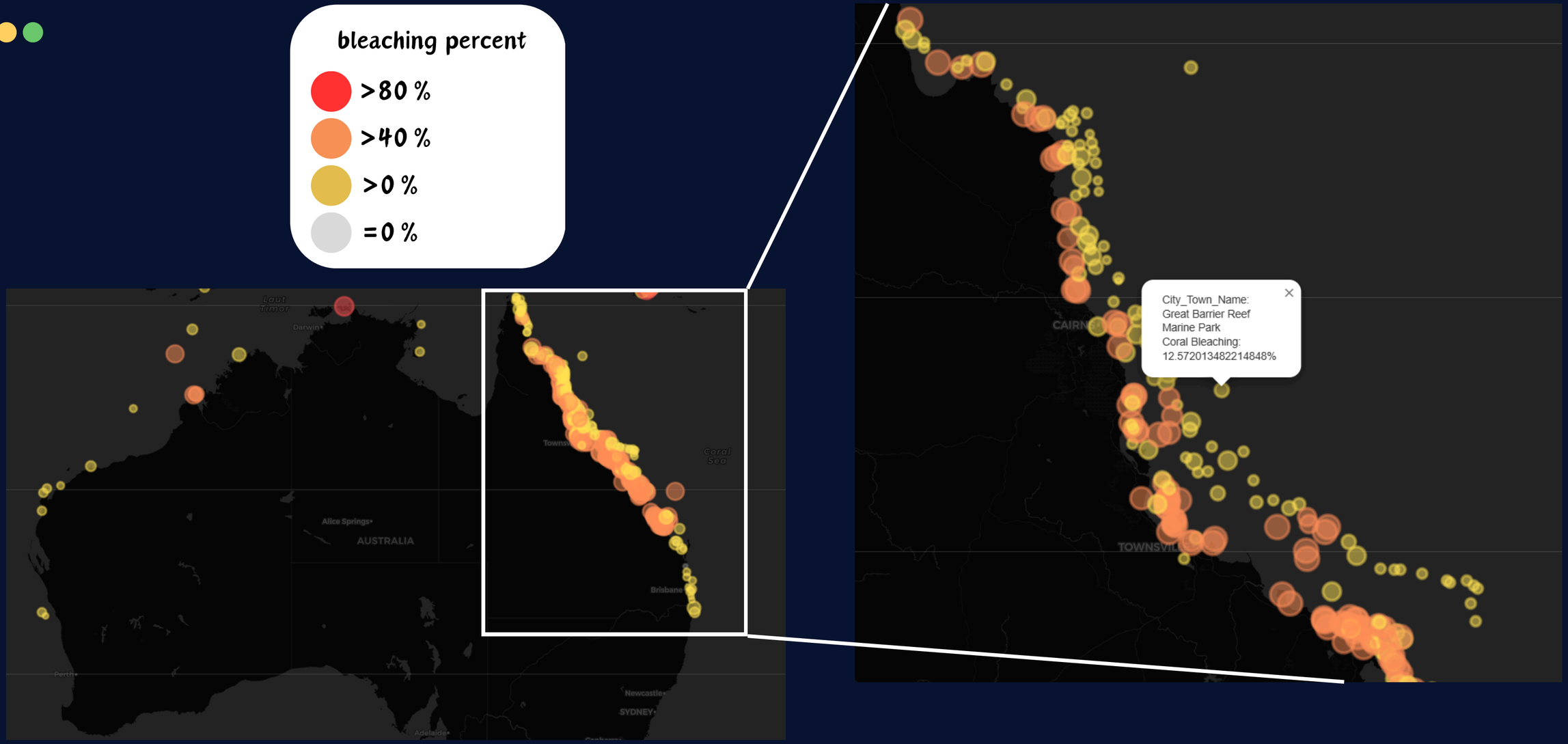
Make a global map of coral bleaching

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import folium

# 載入數據
df = pd.read_csv('global_bleaching_environmental (1).csv', encoding='latin1')

def generate_popup(City_Town_Name, coral):
    return f'''City_Town_Name: {City_Town_Name}<br>Coral Bleaching: {coral}%'''

# 處理數據
df['Percent_Bleaching'] = pd.to_numeric(df['Percent_Bleaching'], errors='coerce')
df['Percent_Bleaching'] = df['Percent_Bleaching'].fillna(df['Percent_Bleaching'].mean())

# 計算每個城鎮的平均漂白百分比
gb = df.groupby('City_Town_Name').mean(numeric_only=True).reset_index()
gb = gb.rename(columns={'Percent_Bleaching': 'PB'})

# 設置顏色和半徑
gb['color'] = ['#FF5151' if x >= 80 else '#FF8F59' if x >= 40 else '#FFE153' if x > 0 else '#7B7B7B' for x in gb['PB']]

min_pb = gb['PB'].min()
max_pb = gb['PB'].max()
base_size = 3
scale_factor = 10

# 計算圓形半徑的函數
def calculate_radius(row):
    # 正規化 PB 值
    normalized_pb = (row['PB'] - min_pb) / (max_pb - min_pb)
    # 計算半徑
    radius = base_size + scale_factor * (normalized_pb ** 0.5)
    return radius

# 為每一行新增一個計算出的半徑列
gb['radius'] = gb.apply(calculate_radius, axis=1)

# 創建地圖，使用第一個城市的經緯度作為地圖中心
latitude_center = df['Latitude_Degrees'].mean()
longitude_center = df['Longitude_Degrees'].mean()

map = folium.Map(
    location=[latitude_center, longitude_center],  # 地圖中心
    zoom_start=8,
    tiles='Cartodb dark_matter',
    width=1500,
    height=900
)

# 將每個城市的圓形標記添加到地圖上
for _, row in gb.iterrows():
    folium.CircleMarker(
        location=[row['Latitude_Degrees'], row['Longitude_Degrees']],  # 使用每個城市的經緯度
        popup=generate_popup(row['City_Town_Name'], row['PB']),
        color=row['color'],       # 外框顏色
        fill=True,                # 是否填滿
        fillColor=row['color'],   # 填充顏色
        opacity=0.5,              # 外框透明度
        fill_opacity=0.5,         # 填充透明度
        radius=row['radius']      # 使用計算出的半徑
    ).add_to(map)
map.save('global_coral_bleaching_map.html')
```



- every site of coral reef bleaching point

result



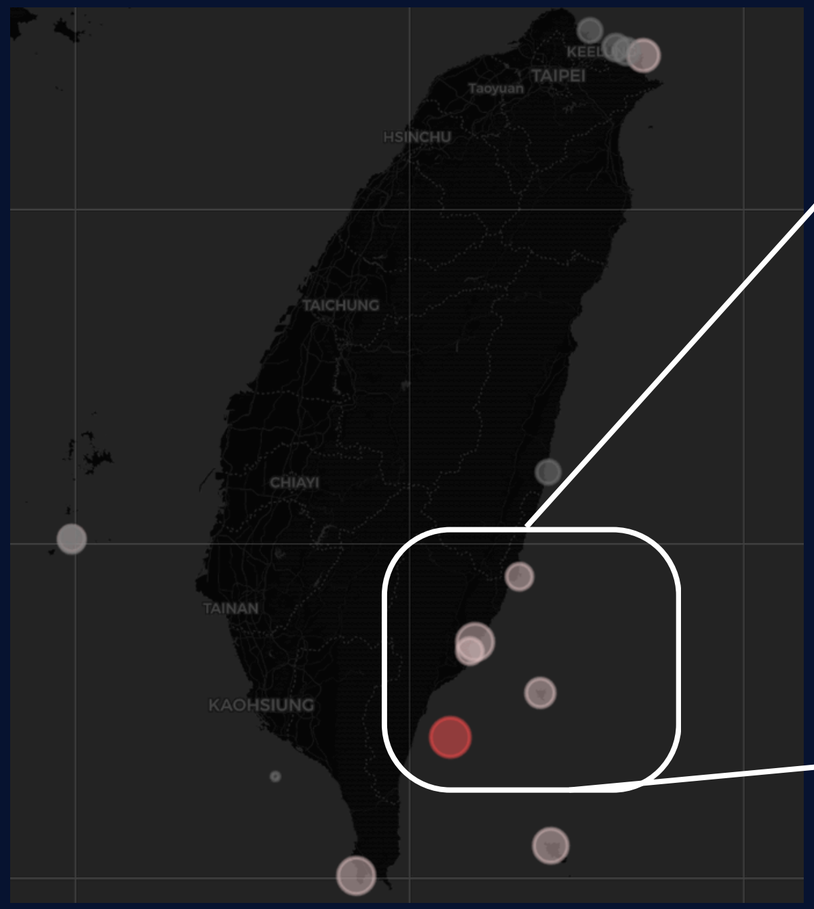
result

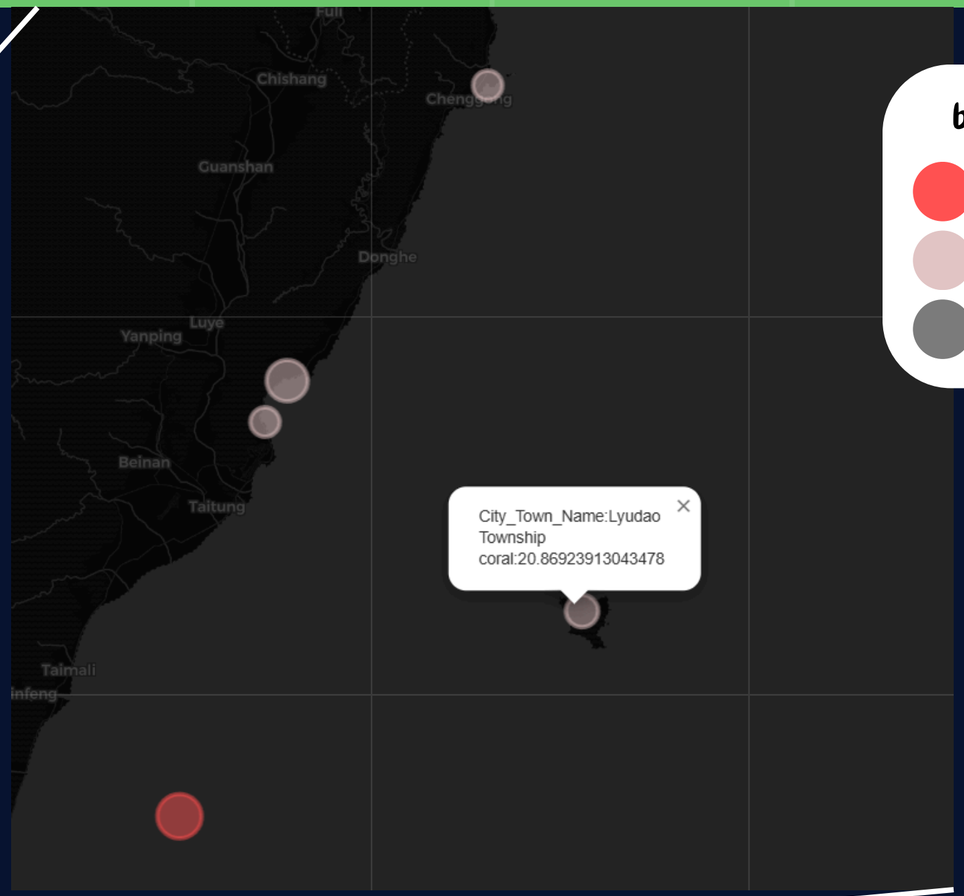
- every site of coral reef bleaching point

Taiwan coral reef cover percent

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import folium
import pandas as pd

df=pd.read_csv('1.csv')

gb=df.groupby(['Latitude_Degrees','Longitude_Degrees']).mean(numeric_only=True).reset_index()

def generate_popup(City_Town_Name, coral):
   return f'''City_Town_Name:{City_Town_Name}<br>coral:{coral}'''
df = pd.read_csv('1.csv')
map = folium.Map(
    location=[.5*(df['Latitude_Degrees'].min()+df['Longitude_Degrees'].max()), .5*(df['Latitude_Degrees'].min()+df['Longitude_Degrees'].max())],
    zoom_start=8,
    tiles='Cartodb dark_matter',
    width=1500,
    height=900
    )
```

```
gb = df.groupby('City_Town_Name').mean(numeric_only=True).reset_index()
gb['PC'] = gb['Percent_Cover']
gb['color'] = ['#FF5151' if x >= 30 else '#E1C4C4' if x >= 20 else '#7B7B7B'for x in gb['PC']]
min_pc = gb['PC'].min()
max_pc = gb['PC'].max()
for _, row in gb.iterrows():
   radius = 15* ((row['PC'] - min_pc) / (max_pc - min_pc)) ** 0.5
   folium.CircleMarker(
   location=[row['Latitude_Degrees'], row['Longitude_Degrees']],
   popup=generate_popup(row['City_Town_Name'],row['PC']),
   color=row['color'],
   fill=True,
   fillColor=row['color'],
   opacity=.5,
   fill_opacity=.5,
   radius =radius
   ).add_to(map)
map.save('Taiwan_percent_cover_map.html')
```





bleaching percent

- >30 %
- >20 %
- <20 %

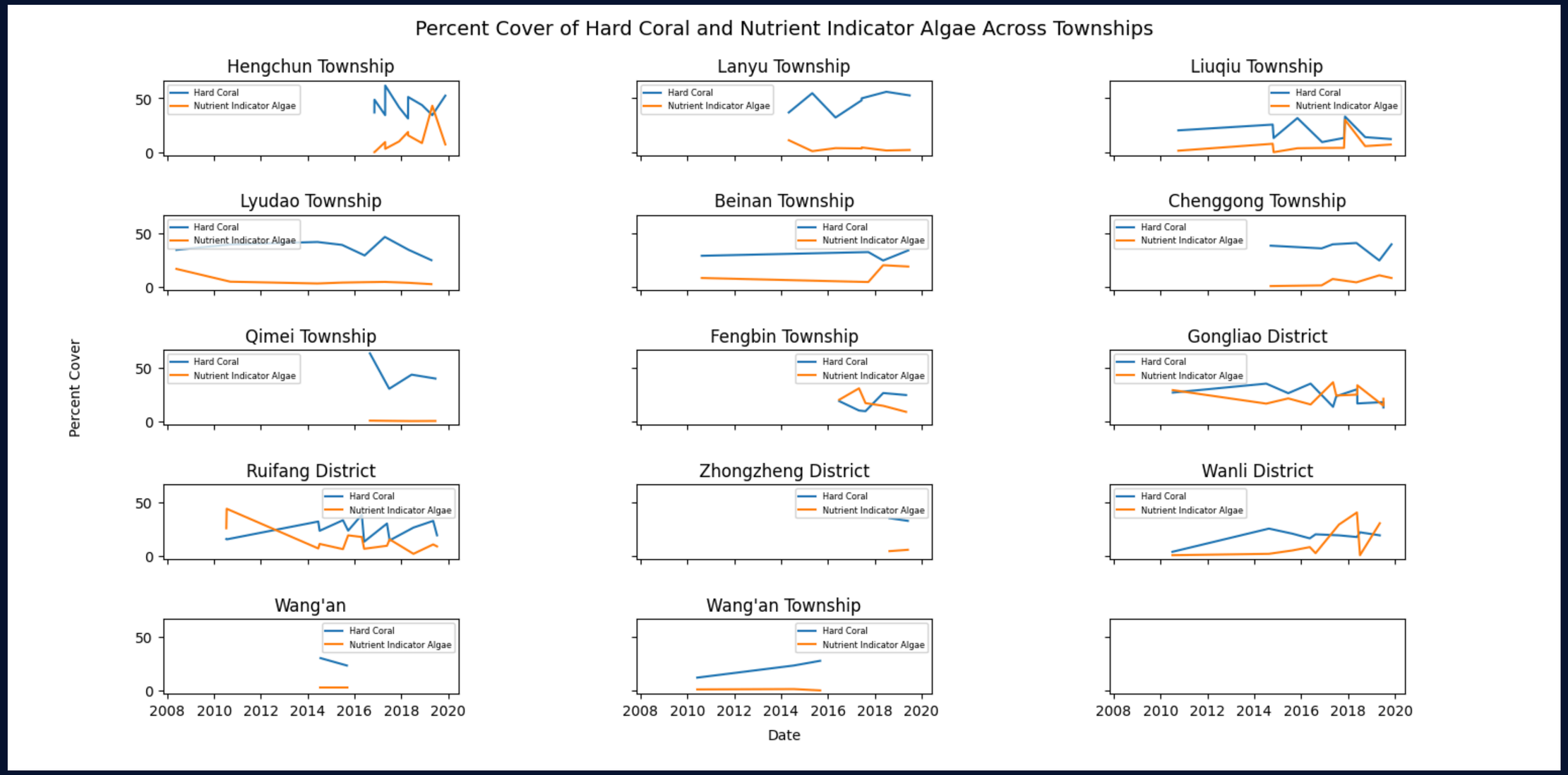
result

- Taiwan coral reef point

Taiwan hard coral and Nutrient Indicator Algae Percent Cover

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from math import isnan
from matplotlib import colorbar, colors, patches

df=pd.read_csv('1.csv')

def process_data(df, city_name, substrates):
    results = {}
    for substrate in substrates:
        subset = df[(df['City_Town_Name'] == city_name) & (df['Substrate_Name'] == substrate)]
        subset['Date'] = pd.to_datetime(subset['Date'], errors='coerce')  # 確保日期正確轉換
        grouped = subset.groupby(subset['Date'].dt.date).mean(numeric_only=True).reset_index()
        results[substrate] = grouped
    return results

# 定義城市和基質類型
city_names = [
    'Hengchun Township', 'Lanyu Township', 'Liuqiu Township',
    'Lyudao Township', 'Beinan Township', 'Chenggong Township',
    'Qimei Township', 'Fengbin Township', 'Gongliao District',
    'Ruifang District', 'Zhongzheng District', 'Wanli District','Wang'an", 'Wang\'an Township'
]
substrates = ['Hard Coral', 'Nutrient Indicator Algae']

# 繪製多城市數據的圖表
fig, axes = plt.subplots(nrows=5, ncols=3, figsize=(18, 16), sharex=True, sharey=True)
axes = axes.flatten()

for idx, city_name in enumerate(city_names):
    ax = axes[idx]
    processed_data = process_data(df, city_name, substrates)
```

```
    # 繪製每種基質類型的曲線
    for substrate, data in processed_data.items():
        ax.plot(data['Date'], data['Percent_Cover'], label=substrate)

    # 設置子圖標題和樣式
    ax.set_title(city_name)
    ax.legend(fontsize=6)
    ax.grid(False)

# 全局設置
fig.suptitle("Percent Cover of Hard Coral and Nutrient Indicator Algae Across Townships", fontsize=14)
fig.text(0.5, 0.04, 'Date', ha='center', fontsize=10)
fig.text(0.04, 0.5, 'Percent Cover', va='center', rotation='vertical', fontsize=10)
plt.subplots_adjust(left=0.1, bottom=0.1, right=0.9, top=0.9, hspace=0.8, wspace=0.6)

# 顯示圖表
plt.show()
fig.savefig('Percent Cover of Hard Coral and Nutrient Indicator Algae Across Townships.png')
```



result

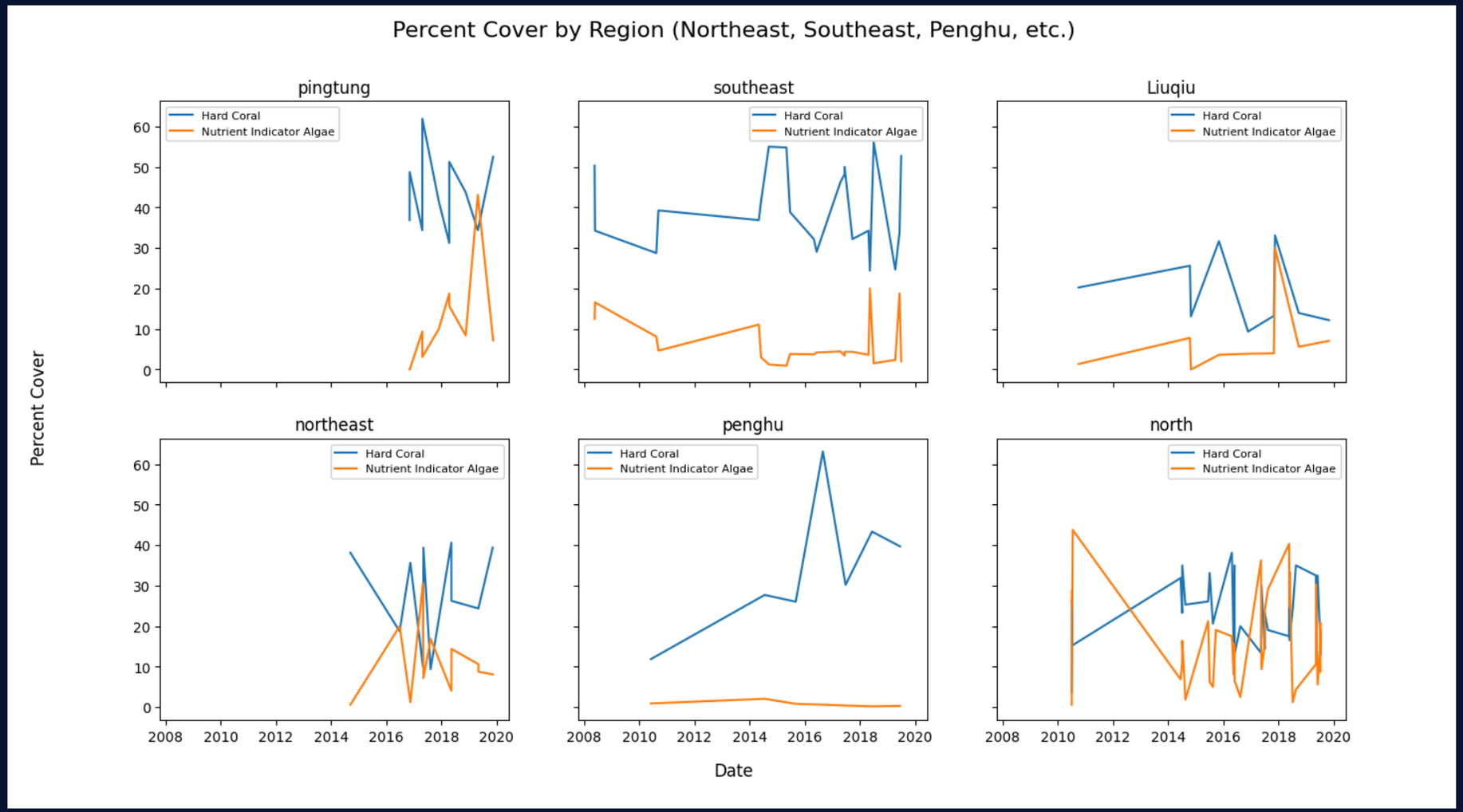
- percent cover of hard coral and nutrient indicator algae across townships

Divided into six regions for comparison

```python
northeast_towns = ['Fengbin Township','Chenggong Township']
southeast_towns = ['Donghe Township','Beinan Township','Lyudao Township','Taimali Township','Lanyu Township']
penghu_towns = ['Qimei Township',"Wang'an", 'Wang\'an Township']
pingtung_county=['Hengchun Township']
north_towns=['Wanli District','Zhongzheng District','Ruifang District','Gongliao District']
Liuqiu_towns=['Liuqiu Township']

def classify_region_by_township(city_name):
    if city_name in northeast_towns:
        return 'northeast'
    elif city_name in southeast_towns:
        return 'southeast'
    elif city_name in penghu_towns:
        return 'penghu'
    elif city_name in pingtung_county:
        return 'pingtung'
    elif city_name in north_towns:
        return 'north'
    else:
        return 'Liuqiu'
```

```python
def process_data_by_region_and_township(df, substrates):
    #根據鄉鎮和基質類型處理數據。

    df['region'] = df['City_Town_Name'].apply(classify_region_by_township)
    results = {}
    for region in df['region'].unique():
        region_data = {}
        for substrate in substrates:
            subset = df[(df['region'] == region) & (df['Substrate_Name'] == substrate)]
            subset['Date'] = pd.to_datetime(subset['Date'], errors='coerce')  # 確保日期正確轉換
            grouped = subset.groupby(subset['Date'].dt.date).mean(numeric_only=True).reset_index()
            region_data[substrate] = grouped
        results[region] = region_data
    return results

# 定義基質類型
substrates = ['Hard Coral', 'Nutrient Indicator Algae']

# 計算每個區域的數據
region_data = process_data_by_region_and_township(df, substrates)

# 繪製多城市數據的圖表
fig, axes = plt.subplots(nrows=2, ncols=3, figsize=(18, 16), sharex=True, sharey=True)
axes = axes.flatten()

for idx, region in enumerate(region_data.keys()):
    ax = axes[idx]
    data = region_data[region]
    for substrate, grouped_data in data.items():
        ax.plot(grouped_data['Date'], grouped_data['Percent_Cover'], label=substrate)
```

```python
# 設置子圖標題和圖例
    ax.set_title(region)
    ax.legend(fontsize=8)
    ax.grid(False)  # 去除格線

# 全局設置
fig.suptitle("Percent Cover by Region (Northeast, Southeast, Penghu, etc.)", fontsize=16)
fig.text(0.5, 0.04, 'Date', ha='center', fontsize=12)
fig.text(0.04, 0.5, 'Percent Cover', va='center', rotation='vertical', fontsize=12)

# 調整布局並顯示圖表
plt.tight_layout(rect=(1,0,0,0))
plt.show()
fig.savefig('Percent Cover by Region (Northeast, Southeast, Penghu, etc.).png')
```



result

- percent cover by region

(north,northeast,southeast,Penghu,Liuqiu,Pingtung)

結果與討論
(result and discussion)

1. The inverse relationship between Nutrient Indicator Algae and Hard Coral Coverage.
2. In the future, studies can be conducted at the same time points to investigate why certain areas exhibit different changes. This could involve exploring events or factors that occurred during those periods.
3. Future studies can also include other oceanic conditions (such as temperature, nutrients, acidity, etc.) to create a correlation chart with coral coverage rates.

The possible reasons for this include the following:

1. Eutrophication caused by land-based pollutants is a major factor contributing to the excessive growth of nutrient-indicator algae. This indicates that corals during this period are subjected to significant pollution, leading to a decrease in their coverage rate.
2. The metabolic processes of algae may release certain chemicals (such as secondary metabolites) that are toxic to coral larvae, further inhibiting coral recovery.